

EXODUS COLLEGE SCHOOL

UGANDA ADVANCED CERTIFICATE OF EDUCATION

SENIOR FIVE MATHEMATICS

CONTINUOUS ASSESSMENT TERM 2 2025

2 HOURS

Attempt any four questions

ITEM ONE

In a certain school, the department of physical sciences wanted to know the relationship between the physics and mathematics performance among PEM and PCM students. The tests were given in both subjects and results of 10 students were recorded as follows.

Student	A	B	C	D	E	F	G	H	I	J
Maths	35	54	80	95	73	73	35	91	83	81
Physics	40	60	75	90	70	75	38	95	75	70

- By taking Maths results as the horizontal axis. Determine the mean point(X, Y).
- Plot a scatter graph and draw a line of best fit.
- calculate the spear man's rank correlation coefficient and comment on the result

ITEM TWO

By using a calculator, a senior five student rounded off values $a = 23.057$ and $b = 8.4657$. To a given number of decimal places.

- State the maximum possible errors in a and b
- Determine the expression for the maximum relative error in $\frac{a}{b}$
- Find the range of values with in which $\frac{(a-b)}{a}$ lies.

ITEM THREE

Box A contains 5 Red and 4 White marbles, **box B** contains 7 Red and 5 White marbles. A box is chosen at random and two marbles are selected from it without replacement. Find the probability that;

- They are of the same colour.
- at least one red marble is selected
- Box b is chosen given that the two marbles selected are of different colours.

ITEM FOUR

A telecommunication company plans to introduce a new minutes bundle on the menu and employs callers to investigate people's opinion on the minutes of the package that will cost shs. 1,200. The callers made a survey on random calls of the customers to gather their opinion and use the average minutes. They collected the data below.

52	36	76	51	62	67	70	50
45	49	54	58	53	74	64	56
50	80	70	57	64	64	43	78
84	71	85	72	78	43	42	75
84	72	69	49	66	42	65	88

- Giving a reason based on the computations using the collected data, suggest the suitable number of minutes to be added on the menu.
- the supervisor wishes to know the number of people who prefer minutes greater than 55. Help him/her to estimate this number of people by using a suitable graph.

ITEM FIVE

In an experiment to demonstrate the effect of forces on a body, a teacher applied the following forces on a solid metallic block; **5N** and **13N** forces acting vertically upwards, **11N** force acting vertically downwards, **3N** and **5N** forces acting horizontally to the left and **14N** acting horizontally to the right. The teacher tasked the students to determine a single force resulting from the effect of these six forces acting on the block.

In his further demonstration he played a video that involved a monster truck at the centre of five (5) small cars, each connected to it using a strong cable. The cars pulled apart with forces of **30N**, **60N**, **100N**, **70N** and **20N** in the directions **270°**, **040°**, **120°**, **330°** and respectively. The teacher also asked the students to use the system of forces above to determine a single force with which the monster truck will resultantly move and the direction it will take.

Task

As a student of mechanics help the students to accomplish the tasks in each of the cases above.

END